TÉCNICO LISBOA

centra 30th Anniversary

ASTROPHYSICS LABORATORY
Observational Astrophysics

I) Basic notions in Observational astrophysics

a) Properties of the astronomical objects

- Magnitude; Absolute magnitude
- Magnitude and distance: ly, parsec
- Temperature and colour
- Size

b) Preparing astronomical observations

- **Visibility of the target:** coordinates and time
- Telescope
- Detector and data acquistion

II) Observations: data acquisition

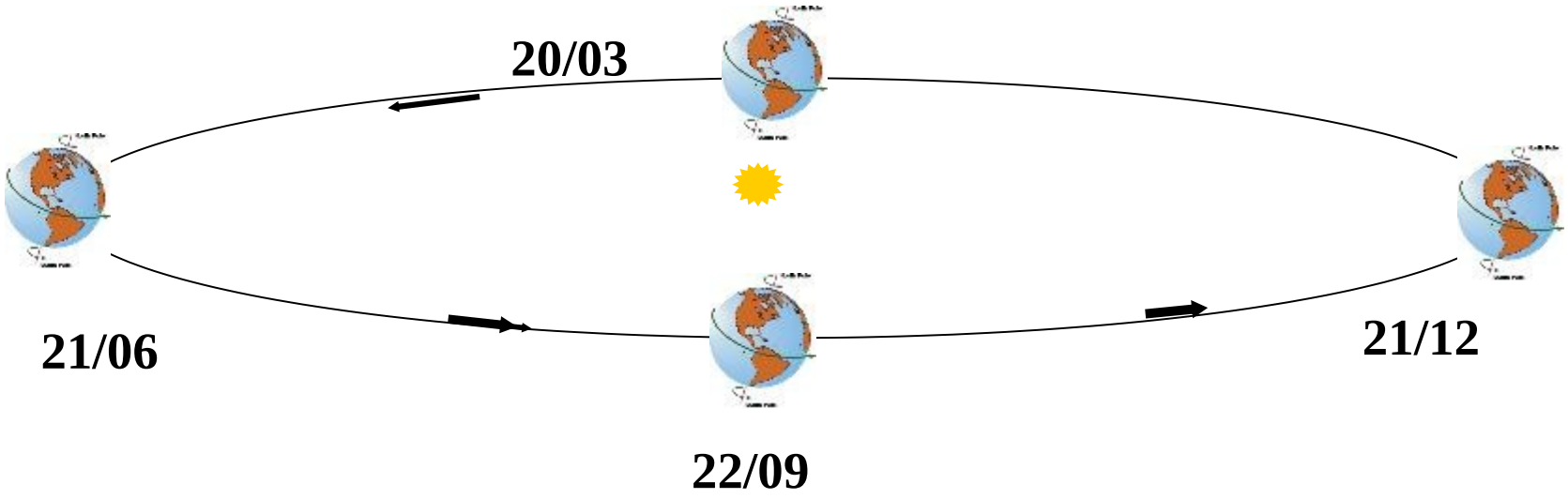
III) Data reduction and analysis

OBSERVATIONS

Coordinates of the observer	Latitude and longitude of the observatory
Date and time	Year, day, hour
Coordinates of the targets	Celestial coordinate system
Exposure time	



OBSERVATIONS





Coordenadas GPS

Latitude: N 38º 46’ 42.03" (38.778342º)

Longitude: W 9º 06’ 49.53" (-9.113758º)

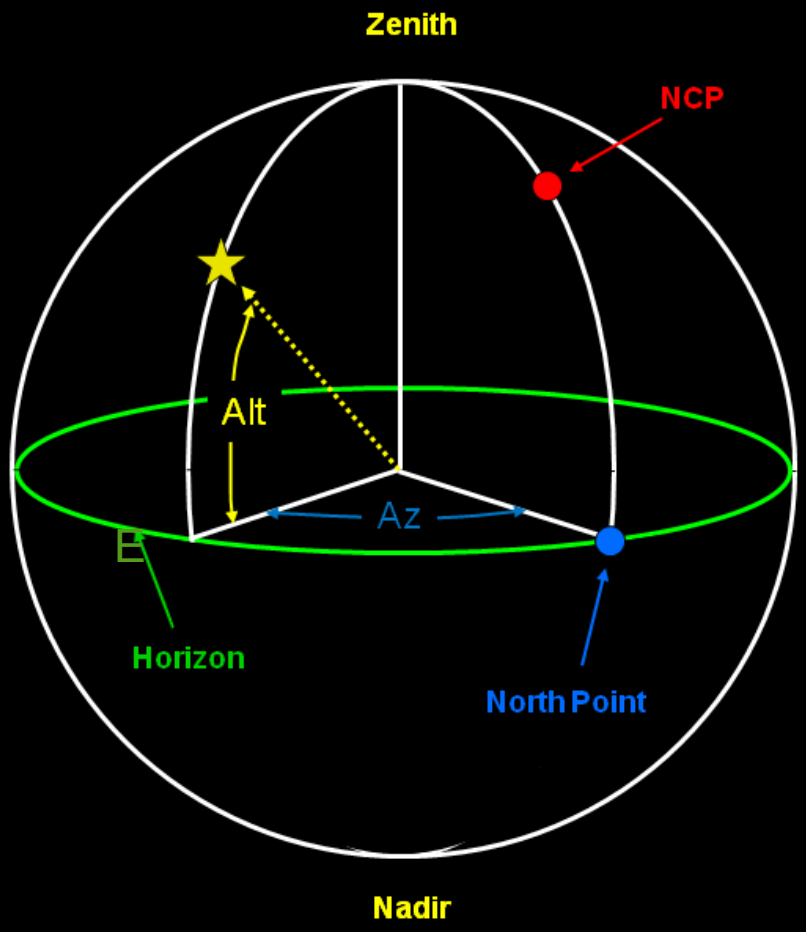
Celestron 14” in Lisbon

Coordinate systems

- Observer coordinate system: Alt and Az
- Equatorial coordinate system: RA and DEC
- Galactic coordinate system: l and b

Observer coordinate system

Altitude and azimuth



Local horizon

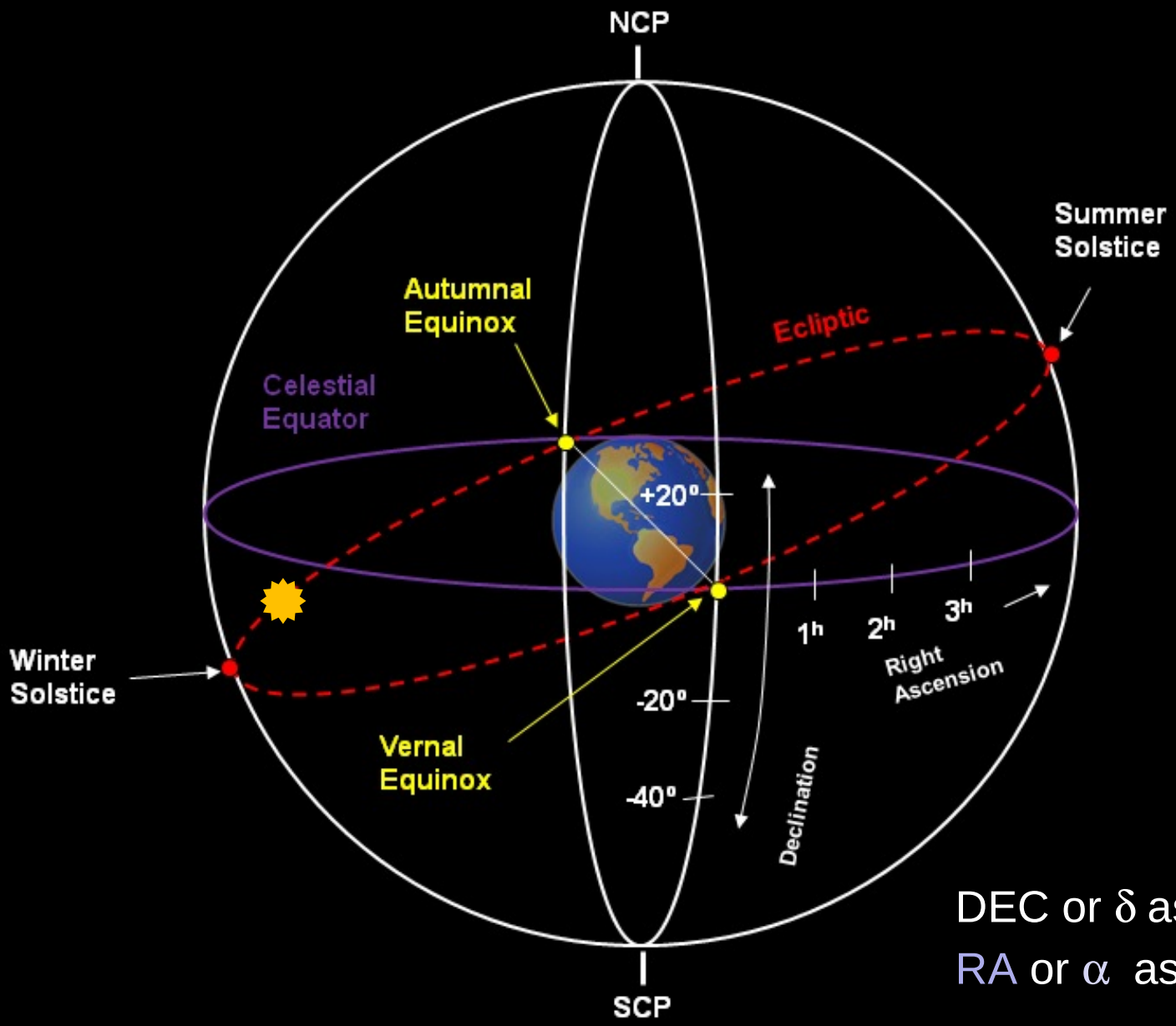
Equatorial coordinate system: RA and DEC



Equatorial coordinate system: RA and DEC



Equatorial coordinate system: RA and DEC



DEC or δ as latitude

RA or α as hour angle

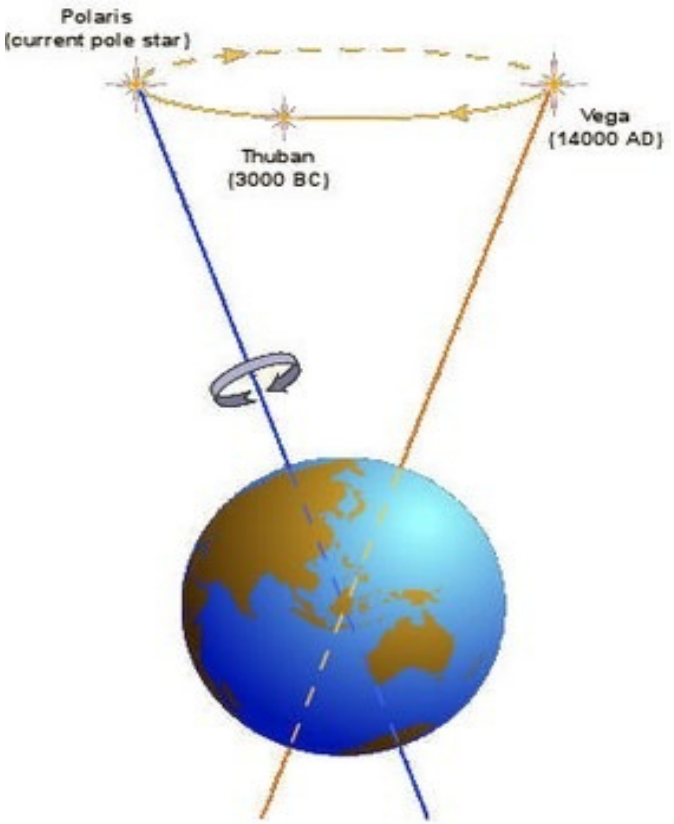
Coordinate systems

Earth precession

Earth precession
the rotational axis changes with a period of 25770 years

The equatorial coordinate system depends on the Earth's rotational axis
coordinates change with time

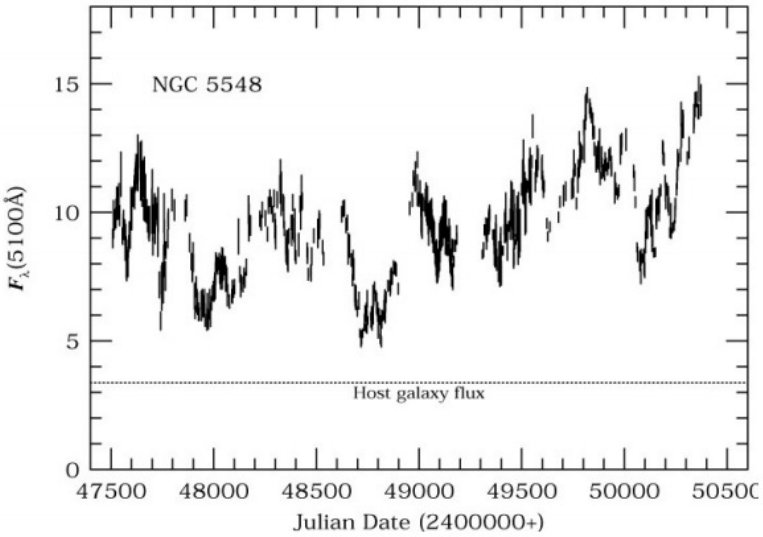
Epoch "J2000": coordinates in 2000 AD.



Time systems: Julian Date

- **Julian date (JD)** system allows to have continuous day counting (no months, years…)
  Number of days since noon, UT on 1st of January 4173 BC
- **Modified Julian Date (MJD)**: Number of days since 17th of November 1858.

$$MJD = JD - 240000.5$$



OBSERVING at the AO-CIGeoE



CIGeoE Telescope
Application for observing time

Period: 2025B

TITLE

ABSTRACT (max. 100 words)

Total time requested (hrs):

Camera requirement ('y' or 'n'):

Moon conditions ('d','g','n'):

Filters required (BVR):

Investigators

Scientific Rationale (max. 450 words, max. 2 figures)

References

Targets

NAME	α (J2000)	δ (J2000)	Time on target	Mag	Additional info

Observing plan suggestion and time justification (max 150 words, figures allowed)

Panel Evaluation (leave blank):
Abstract (20%):

Project Ideas

Nebulae

Scientific questions:

- What is a nebula? What is its chemical composition? How long ago has it exploded?

Skills:

- Measure luminosity profile and integrated luminosity of a nebula
- Obtain a color-composite image

Tips:

- For the exposure time calculation, nebulae are extended sources
- Make sure (most of) the nebula fits in the field of view. Mosaics are possible but challenging.

Project Ideas

Stellar Clusters

Scientific questions:

- What is a stellar cluster? What is its age? What is the turnoff point?

Skills:

- Obtain magnitudes and colors of each star.
- Plot the stars in a Hertzsprung-Russell diagram
- Obtain a color-composite image

Tips:

- *To calculate exposure times, use faint stars in stellar cluster. See the member of a cluster with SIMBAD: http://simbad.u-strasbg.fr/simbad/*
- *Do not pick stellar clusters that are too crowded so that you can resolve more stars (and do photometry)*
- *Make sure (most of) your stellar cluster fits in the field of view*

Project Ideas

Individual Stars

Scientific questions:

- What are the types of some of the stars we see in the night sky? How will they evolve?

Skills:

- Calculate the color and temperature of stars.
- Plot the stars in a Hertzsprung-Russell diagram

Tips:

- Be careful when selecting stars to ensure that they do not saturate the CCD.

Planning an observation

Ephemeris: trajectories of objects in the sky

Visibility: of an object from a certain place during a certain time.

Tools

- Stellarium: home planetarium (Mac, Win, Linux)
- SkyMap: mobile sky tool
- Visibility tools: https://airmass.org

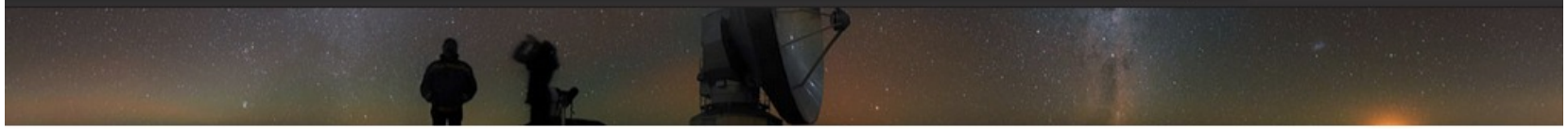
Planning an observation

- Plan the observations at the CiGeoE telescope
- Order the objects to observe, from sunset to 23h Lisbon time.
- Each object should be observed for 0.5h and count with 10min to move the telescope to the new position.

Lisbon: -9.1393W 38.7223 10 N
Objects

- Moon
- Saturn (?)
- Jupiter (?)
- Albireo
- Mizar & Algol
- Antares, Deneb
- Orion Nebula (M42)
- Andromeda (M31)
- Hercules Cluster (M13)
- Ring Nebula (M57)
- Dumbbell Nebula (M27)
- Crab Nebula

https://airmass.org/chart/obsid:Geolocated position/el:94/tz:Europe-Lisbon/lat:38.75852



Object [?]

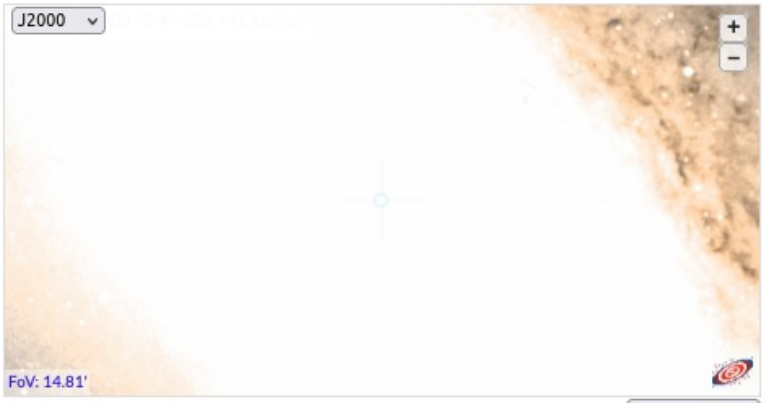
Deep sky object | Solar system object | Multiple objects

aNDROMEDA resolve

00 42 44.33 41 16 07.50 get from map

advanced options...

Object	RA	dec	[clear all]
M13	16 41 41.63	36 27 40.75	[SIMBAD]
p:Saturn	23 59 46.86	-02 44 56.43	[JPL]
M57	18 53 35.10	33 01 44.88	[SIMBAD]
M27	19 59 36.36	22 43 16.31	[SIMBAD]
aNDROMEDA	00 42 44.33	41 16 07.50	[SIMBAD]



identify objects

Observatory [?]

Geolocated position

38 45 30.68

-09 10 08.34

94

Europe/Lisbon

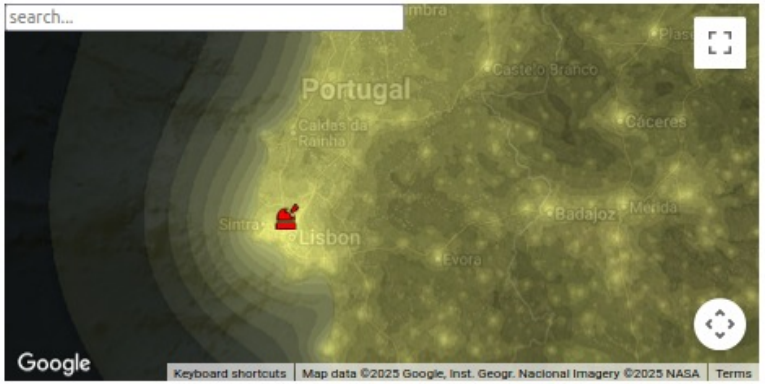
set to my location | set to nearest observatory

Weather at 9:25:33 AM local time: few clouds

		%	
18.5°C	10mph ↓	77%	20%

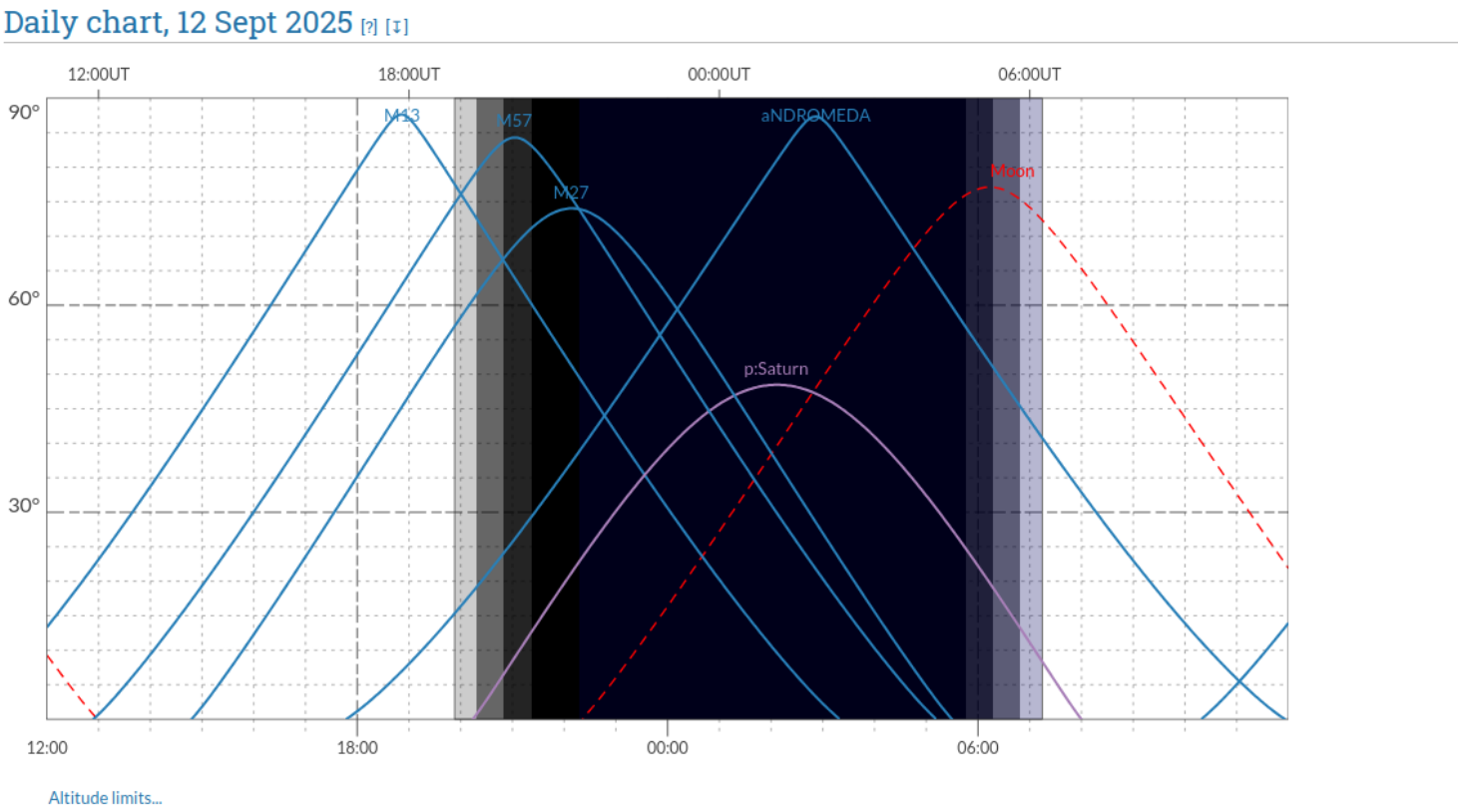
hide forecast

Thursday	broken clouds	25.7°C	17mph ↓	54%	66%
Friday	overcast clouds	27.7°C	14mph ↘	33%	100%
Saturday	scattered clouds	24.5°C	11mph →	42%	33%



aurorae clouds light pollution night

Planning an observation: VISIBILITY



website for the LA2025

TÉCNICO LISBOA

Laboratório de Astrofísica

Useful material:

- Visibility tools:
  https://airmass.org/
- Exposure time calculator:
  https://www.mirametrics.com/sn_calculator_mvn.php
- Field of view calculator:
  https://www.12dstring.me.uk/fovcalc.php
- Telescope and camera info:
  https://www.qxccd.com/art?id=321
- Light pollution:
  https://www.lightpollutionmap.info/#zoom=4&lat=5759860&lon=1619364&layers=B0FFFFFTFFFFF
- Moon brightness:
  https://www.mooncalc.org

- stellarium: home planetarium for planning observations

Software:

The following software is recommended. You can install them beforehand on your computers. We may use jupyter notebooks (or Google Collabs) found in this folder.

- python3: astropy, photutils, astroplan

- ds9: for visualizaing FITS files

- astrometry.net: for doing astrometry

Other useful software: AstroImageJ

Why would you get time to access a telescope?

Observing proposal

WHY WRITE AN OBSERVING PROPOSAL?

- You have a good scientific question/problem that you want to answer/solve

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WHY NOT SIMPLY OBSERVE IT?

- Strong competition for limited resources

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  Oversubscription rate = time requested/time available

Telescope	Oversubscription
Small telescopes (0.5-2m)	0.5 - 2
Middle telescopes (4m)	2-3
Large telescopes (8m)	5-6
Space telescopes	6-13

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- Astronomical instrumentation is expensive

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Telescope	Cost [€/hour]
VLBA (radio)	740
Calar Alto (2.2m)	360
ESO NTT (3.5m)	1200
ESO VLT (8.0m)	7280
HST (2.4m space)	40000
JWST (6.5m space)	115000

Observing proposal

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HOW IS THE PROPOSAL PROCESS?

- Observatories/agencies send a call for proposals (twice a yr)
- Panel committee evaluates proposals for: a) scientific case and b) technical feasibility and they are ranked (Not all members are specialists!)
- Rejection/acceptance is sent and observing priority (A,B,C)

Observing proposal

HOW TO WRITE AN OBSERVING PROPOSAL?

- Investigate your scientific idea and convince the others about its importance and relevance
- Understand your instrument and the feasibility of the study (visibility, S/N, etc)

Observing proposal

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HOW IS THE STRUCTURE OF THE PROPOSAL?

- Abstract: short summary
- Scientific justification
- Technical feasibility
  - Targets and suggested observing dates

Proposals for the AO-CIGeoE

1. Make groups of max. 3 people and pick a scientific observing project that interests you

2. Find targets that are visible during the week of Sep. 11-26th 2024 from 20h-23h.

3. Write a small proposal (deadline: 11th Sep, 4pm:00) with abstract, scientific case and time justification

4. For time justification calculate exposure times and take into account overheads